

Day 14: Advanced DAX Filter Modifiers

📌 Definition to Remember

Filter modifier functions (**ALL** , **ALLSELECTED** , **REMOVEFILTERS** , **KEEPFILTERS** , **FILTER**) work inside **CALCULATE()** to explicitly control which dimension filters are removed, preserved, or dynamically evaluated.

1. **ALL()** vs **REMOVEFILTERS()** — Calculating % of Total

ALL() removes all filters from a table or column, returning all rows:

```
// Grand Total Sales (Ignores all product/category slicers)
Grand Total Sales =
CALCULATE(
    [Total Sales],
    ALL(Dim_Product)
)

// Contribution % to Grand Total
Product Sales % of Total =
DIVIDE([Total Sales], [Grand Total Sales], 0)
```

2. **ALLSELECTED()** — Dynamic Visual Contribution %

ALLSELECTED() removes visual internal axis filters while **preserving external user slicer selections**:

```
Sales % of Sliced Selection =
DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales], ALLSELECTED(Dim_Product[Category])),
    0
)
```

3. The **FILTER()** Iterator Function

FILTER(Table, Condition) iterates through a table row-by-row to apply complex measure-based filtering:

```
Sales from High Margin Products =
CALCULATE(
    [Total Sales],
    FILTER(
        Dim_Product,
        [Profit Margin %] > 0.25 // Filtering by a dynamic measure
    )
)
```

★ Must-Master Skills & Interview Points

1. Differentiate between `ALL()` (ignores everything) and `ALLSELECTED()` (respects user slicers).
2. Why `REMOVEFILTERS()` is the modern, semantic replacement for `ALL()` as a filter modifier.
3. Performance tip: Do NOT wrap simple column comparisons inside `FILTER()` unless filtering by a dynamic measure.

⚡ Quick Recall

`ALL` (Clear all filters) + `ALLSELECTED` (Clear axis but keep slicers) + `FILTER` (Table iterator for complex criteria)